GroveNFC I2C Register List

1 GroveNFC I2C slave address: 0x48

2 I2cSysReg – 16bits register

2.1 I2cSysReg_HardwareVersion 0x0000

Bit	Description	R/W	Default
15-0	Hardware version	RO	0x6123

2.2 I2cSysReg_FirmwareVersion 0x0002

Bit	Description	R/W	Default
15-0	Firmware version	RO	-

2.3 I2cSysReg_Mode 0x0004

Bit	Description	R/W	Default
15-8	Work mode: 00- None 01- Reader 02- Tag 03- Mifare 1 reader	W	0x00
7-0	Valid only in Tag mode 00- None 01- Ntag213 02- Ntag215 03- Ntag216 04- Miafre 1 4B1K 05- Mifare 1 7B1K 06- Mifare 1 4B4K 07- Mifare 1 7B4K 08- Nxp iSLI 09- Nxp iSLIX 0A- Nxp iSLX2 0B- Nxp iSLI_L 0C- Ti IT_HF_I_PLUS 20- China Second-Generation ID Card(Only UID)	W	

2.4 I2cSysReg_TagAddr 0x0006

Bit	Description	R/W	Default
15-0	Address of tag’s data; Must be a multiple of 4096 bytes	W	-

2.5 I2cSysReg_RFConfig 0x0008

Bit	Description	R/W	Default
15-8	Reader config 00- None 01- ISO14A_106K 02- ISO14A_212K 03- ISO14A_424K 04- ISO14A_848K 05- ISO14B_106K 06- ISO14B_212K 07- ISO14B_424K 08- ISO14B_848K 09- FELICA_212K 0A- FELICA_424K 0B- ISO15_26K_ASK100 0C- ISO15_26K_ASK10	W	-
7-0	Tag config 00- None 01- ISO14443A_106K 02- ISO14443A_212K 03- ISO14443A_424K 04- ISO14443A_848K 05- ISO14443B_106K 06- ISO14443B_212K 07- ISO14443B_424K 08- ISO14443B_848K 09- FELICA_212K 0A- FELICA_424K 0B- ISO15693_26K_ASK(single subcarrier) 0C- ISO15693_26K_FSK(two subcarriers)	W	-

2.6 I2cSysReg_FWI 0x000A

Bit	Description	R/W	Default
15-0	FDT for ISO14A; FWI for ISO14B	W	-

2.7 I2cSysReg_TxCrcEn 0x000C

Bit	Description	R/W	Default
15-0	Tx data auto appended crc; 1-Enable 0-Disable Bit0- 14A_ENABLE Bit1- 14B_ENABLE Bit2- FELICA_ENABLE Bit3- ISO15_ENABLE	W	0x0000, Disable crc

2.8 I2cSysReg_RxCrcEn 0x000E

Bit	Description	R/W	Default
15-0	Rx data include crc; 1-Enable 0-Disable Bit0- 14A_ENABLE Bit1- 14B_ENABLE Bit2- FELICA_ENABLE Bit3- ISO15_ENABLE	W	0x0000, Disable crc

2.9 I2cSysReg_NFCStatus 0x0010

Bit	Description	R/W	Default
15-0	NFC status; 1-Active 0-Inactive Bit0- RECV_DONE Bit8- MIFARE1_WRITE_ERR Bit9- MIFARE1_WRITE_OK Bit10- MIFARE1_AUTH_ERR Bit11- MIFARE1_AUTH_OK Bit12- RECV_BIT_ERR Bit13- RECV_CRC_ERR Bit14- RECV_TIMEOUT Bit15- BUSY	R	0x0000

2.10 I2cSysReg_RxLen 0x0012

Bit	Description	R/W	Default
15-0	Rx data length	R	-

3 I2cMiscReg – 8bits register

3.1 I2cMiscReg_RFOn 0x0020

Bit	Description	R/W	Default
7-1	Reserved		
0	1-ON 0-Off	W	0

3.2 I2cMiscReg_TxLastBit 0x0021

Bit	Description	R/W	Default
7-0	0: All bits of last byte are transmitted 1-7: Number of bits within last byte to be transmitted	W	0

3.3 I2cMiscReg_Thru 0x0022

Bit	Description	R/W	Default
7-1	Reserved		-
0	0- Data will be checked bit-parity or/and crc 1- Data thru mode	W	0

3.4 I2cMiscReg_EGT 0x0023

Bit	Description	R/W	Default

7-0	Only valid in ISO14B	W	0

3.5 I2cMiscReg_Slot 0x0024

Bit	Description	R/W	Default
7-0	Only valid in Felica	R	0

3.6 I2cMiscReg_EpWrite 0x0029

Bit	Description	R/W	Default
7-0	1- Write 2- Read	W	0

4 I2cDataReg 0x0100

Receive and transmit data address.

5 I2cEpromReg 0x1000